Stochastic Calculus

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Abstract

We aim to deconstruct the formulation of Itô's lemma and analyze its application to the approximation of stochastic processes.

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1 Motivation

The stochastic calculus provides the foundational framework for a rigorous analysis of investment decisions in environments under uncertainty. Within this framework, our objective is to construct approximations of stochastic processes by utilizing Itô calculus as the principal analytical instrument.

2 Introduction

Itô's Isometry and Itô's lemma are fundamental instruments in stochastic calculus, playing roles analogous to the Riemann integral and the chain rule in classical differential calculus. In deterministic analysis, integration and the chain rule are central to the solution of ordinary differential equations (ODEs). Similarly, in the stochastic setting, Itô's Isometry and Itô's lemma provide the corresponding framework for solving stochastic integral and differential equations.

Itô's Isometry defines integration with respect to a stochastic process, most commonly a Brownian motion (Wiener process). The classical Riemann–Stieltjes integral (Wiki Contributors,) is generally not applicable in this context because Brownian paths are almost surely nowhere differentiable and exhibit fractal-like behavior, with unbounded variation on every time interval. As a consequence, the standard Riemann–Stieltjes construction fails, and a probabilistic (mean-square) limiting procedure is required to define the integral.

Itô's lemma serves as the stochastic analogue of the chain rule, modified to account for the randomness and the non-vanishing quadratic variation of Brownian motion. Specifically, the lemma incorporates an additional second-order term proportional to the quadratic variation of the driving process. This correction term reflects the fact that, although the quadratic variation of Brownian motion over a time interval is non-zero (and in fact equal to the length of the interval almost surely), its expectation is deterministic and scales linearly with the length of the time interval. Together, the Itô integral and Itô's lemma form the core calculus used to analyse and solve SDEs.

Consider the ODE

$$dX_t = a(X_t)dt$$

Then, the integral form is

$$X_t - X_0 = \int_0^t a(X_s)ds$$

The chain rule is represented as a function of the solution $f(X_t)$, which gives

$$\begin{aligned} df(X_t) &= f'(X_t)dX_t \\ &= f'(X_t)a(X_t)dt \end{aligned}$$

The integral form of the differential relation is

$$f(X_t) = f(X_0) + \int_0^t f'(X_s)a(X_s)dsf$$

The expression using differentials and the chain rule is equivalent to the formula involving integrals, in which all the terms are defined. The rigorous integral form does not replace the differential formula. The differential equation shows how the dynamics evolve. The integral form provides confidence that the informal reasoning makes sense.

The Brownian motion is denoted by W_t and the standard Brownian motion is when $W_0 = 0$, $E[W_t] = 0$, and the variance is $Var(W_t) = t$

The Ito integral is a stochastic process

$$X_t = \int_0^t a_s dW_s$$

This is equivalent to the indefinite integral in ODE. The integrand is another random process a_s that must be non-anticipating, meaning only the most current information is used.

The SDE is a dynamic model of the form

$$dX_t = a(X_t)dt + b(X_t)dW_t \tag{1}$$

This equation can be interpreted in two ways:

A. X_t is a stochastic process with infinitesimal mean and variance

$$\mathbb{E}\big[dX_t|X_{[0,t]}\big] = 0 \tag{2}$$

$$Var\big[dX_t|X_{[0,t]}\big] = b^2(X_t)dt \tag{3}$$

This expression follows the informal SDE in (1) using the basic properties of the Brownian motion

$$\mathbb{E}[dW_t] = 0$$
$$Var[dW_t] = dt$$

From the defining properties of Brownian motion, the increment over a small time step satisfies $\Delta W_t \sim \mathcal{N}(0, \Delta t)$. For a normally distributed random variable $X \sim \mathcal{N}(\mu, \sigma^2)$, the expectation is $\mathbb{E}[X] = \mu$; in the present case, this implies that the mean of ΔW_t is zero. Passing to the infinitesimal limit $\Delta t \to dt$, we obtain

$$\mathbb{E}[dW_t] = 0.$$

Furthermore, assuming the initial condition $W_0 = 0$, for any $t > s$ the increment $W_t - W_s$ is independent of the past and, by the stationary and Gaussian increment properties of Brownian motion, it follows that

$$W_t - W_s \sim \mathcal{N}(0, t - s)$$
$$Var[dW_t] = dt$$

B. Take the integral form of (1)

$$X_T - X_t = \int_t^T \alpha(X_s)ds + \int_t^T \sigma(X_s)dW_s$$

where the first term is an ordinary Riemann integral and the other is a Brownian process.

3 Ito's Isometry

Itô's Isometry is a property used when taking the expected value of a squared stochastic integral. It states that the expected value of the squared random integral is exactly equal to the expected value of the standard time integral of the squared integrand. The primary input of this variance tool is a squared stochastic integral inside an expectation operator $\mathbb{E}\left[(\int \cdots dW)^2\right]$, resulting to a standard non-stochastic integral $\int \cdots ds$ that can be evaluated using ordinary calculus. We normally can't predict the exact path of the random variable, but we can precisely measure the range of uncertainty, which is the variance. Because of this, the isometry is used to prove Picard iterations converge by bounding the variance of the errors.

The solution depends on the specific form of the drift, $\alpha(X_s)$, and diffusion, $\sigma(X_s)$, the coefficients being linear or non-linear. The Fokker-Planck (Wiki Contributors,) partial differential equation (PDE) can be used to check if these coefficients are linear. Then apply either the integrating factor method or by using Ito's lemma.

We approximate a stochastic process $f(t)$ with a sequence of simple functions, $\{f_0(t), \cdots, f_n(t)\}$ where each function is constant over the specific Δt partitioned along $[0, T]$ time horizon. We define the time intervals as $(t_0 < t_1), (t_1 < t_2), \cdots, (t_{n-1} < t_{n=T})$. As $n \to \infty$ the time interval , $\Delta t \to 0$. With a simple function, we approximate the integral with respect to time interval as

$$\int_0^T f_n(t)dt \approx \sum_{k=0}^{n-1} f(t_k)\Delta t_k \qquad | \ \Delta t_k < (t_{k+1} - t_k)$$

The integral for the same function with respect to the Brownian process W_t

$$\int_0^T f_n(t)dW_t = \sum_{k=0}^{n-1} f(t_k)dW_t \int_{t_k}^{t_{k+1}} dW_t$$

Using the property, $\int_{t_k}^{t_{k+1}} dW_t = \left(W_{t_{k+1}} - W_{t_k}\right) = \Delta W_k$

$$\int_0^T f_n(t)dW_t = \sum_{k=0}^{n-1} f(t_k)\Delta W_k$$

The sequence of sums converges to a random variable, and for any small error $\epsilon > 0$ and $\delta > 0$, the probability that the difference between the sum and limit is greater than ϵ goes to zero as $n \to 0$. Similarly, the expected square difference between the sums and the limit goes to zero. This is established using the **Ito isometry**

$$\mathbb{E}\left[\left(\int_0^T dW_t\right)^2\right] = \mathbb{E}\left[\int_0^T f(t)^2 dt\right] \tag{4}$$

This property shows that the variance of the Ito integral is equal to the expected value of the integral of the square of the integrand. It is instrumental in proving the convergence of the sum to the Ito integral in mean square.

Example 1

Consider the integral:

$$I_t = \int_0^T f(t, W_t)dW_t \tag{5}$$

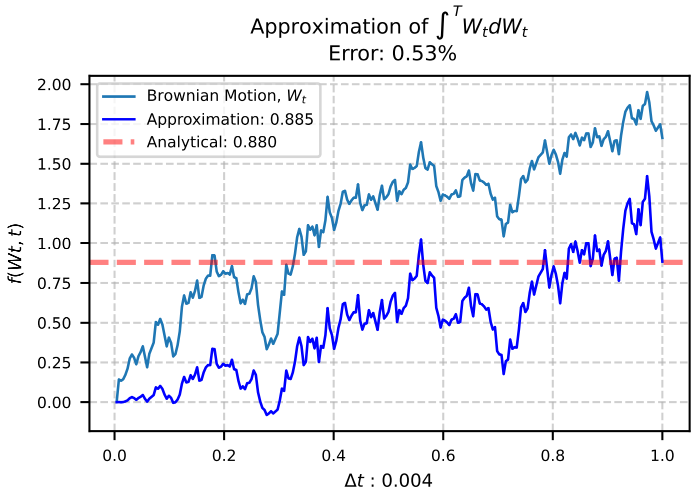

Figure 1: Approximation of Ito integral

We approximate the solution using discretized sum over a partition between $[0, T]$

$$I_t^{(n)} = \sum_{i=0}^{n-1} f(t_i, W_{t_i})(W_{t_{i+1} - W_{t_i}})$$

where dWt is $(W_{t_{i+1} - W_{t_i}}) \sim \mathcal{N}(0, \Delta t)$.

The solution to the integral using Ito's lemma is:

$$\begin{aligned}\int_0^T f(t, W_t)dW_t &= \int_0^T W_t dW_t \\ &= \frac{1}{2}W_t^2 - \frac{1}{2}T\end{aligned}$$

The numerical approximation used the Ito integral

$$\int_0^T H_t dW_t = L^2 - \lim_{n\to\infty} \int_0^T H_t(W_{t_{i+1}} - W_{t_i})$$

where H_t is adapted to the filtration generated by the Brownian process and that the integrand must be evaluated on the left endpoint of t_i. We simulate this solution and is shown in Fig. (1)

Example 2

Consider an integral with a separate integrand $g(t)$

$$\int_0^T g(t)dW_t \tag{6}$$

where $g(t)$ is a process defined by the SDE

$$dX_t = \mu dt + \sigma dW_t$$

Let $\mu = 1$ and $\sigma = 0.5$, this makes $dX_t = 1dt + 0.5dW_t$ an arithmetic Brownian motion with drift.

The integrand $g(t)$ is a process of X_t, therefore, we will calculate $\int_0^T X_t dW_t$, using Ito's lemma to $Y_t = f(X_t) = X_t^2$.

The solution to the integral using Itô's lemma is:

$$\int_0^T X_t \, dW_t = X_T^2 - X_0^2 - 2\int_0^T X_t \, dt - 0.25T$$

This formula gives us a way to calculate the value of the Itô integral if we know X_0, X_T, and the integral of X_t over time. We simulate the given SDE using the values $\mu = 1.0$ and $\sigma = 0.5$. We compare the exact solution using Ito's lemma versus the numerical simulated path as shown in Figure (2)

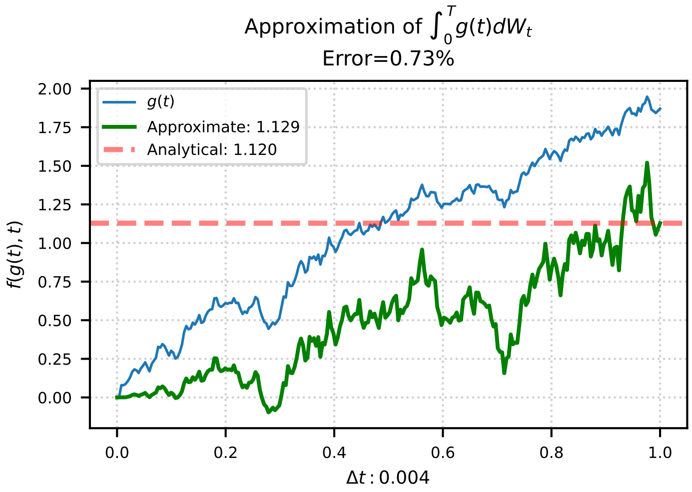

Figure 2: Approximation of $\int_0^T g(t)dW_t$

4 Ito's Lemma

Given a known stochastic process X_t and a smooth function $f(t, X_t)$, Ito's lemma calculates the total differential $df(t, X_t)$. The result is a new SDE that retains the random part W_t and describes how the value of $f(t, X_t)$ changes over time.
Consider a stochastic process $f(X_t, t)$, we derive Ito's lemma using the properties of the Ito integral and Taylor series expansion. The X_t process is defined by the following SDE:

$$dX_t = \mu(X_t, t)dt + \sigma(X_t, t)dW_t \tag{7}$$

We treat the drift and diffusion coefficients as constants μ and σ, respectively. We are interested in the behavior of a certain function $Y_t = f(X_t, t)$. In practical application, we would want to know how returns (Y_t) of a trading strategy will evolve given a price function (X_t) as a stochastic process parameter.

Moreover, we define ΔY over a small time interval from t to $(t + \Delta t)$

$$\Delta Y = Y_{t+\Delta t} - Y_t = f(X_{t+\Delta t}, t + \Delta t) - f(X_t, t)$$

We use the Taylor series expansion of $f(\cdots)$ around the point (X_t, t). For a function of two variables, the Taylor expansion to the second order, ignoring the higher-order derivatives

$$f(X_{t+\Delta t}, t + \Delta t) \approx f(X_t, t) + \frac{\partial f}{\partial x}(X_t, t)\Delta X + \frac{\partial f}{\partial t}(X_t, t)\Delta t + \frac{\partial^2 f}{\partial x^2}(X_t, t)(\Delta X)^2 + \cdots$$

Therefore,

$$\Delta Y \approx \frac{\partial f}{\partial x}\Delta X + \frac{\partial f}{\partial t}\Delta t + \frac{\partial^2 f}{\partial x^2}(\Delta X)^2$$

We substitute the differential of the SDE for ΔX : $dX_t = \mu dt + \sigma dW_t$. For a small interval, $\Delta X =\approx \mu\Delta t + \sigma\Delta W$

Expanding $(\Delta X)^2$:

$$(\Delta X)^2 \approx (\mu\Delta t + \sigma\Delta W)^2$$
$$(\Delta X)^2 \approx (\mu\Delta t)^2 + 2(\mu\Delta t)(\sigma\Delta W) + (\sigma\Delta W)^2$$
$$(\Delta X)^2 \approx \mu^2(\Delta t)^2 + 2\mu\sigma\Delta t\Delta W + \sigma^2(\Delta W)^2$$

We need to determine the behavior of $(\Delta t)^2$, $\Delta t\Delta W$, and $(\Delta W)^2$ as $\Delta t \to 0$.

From the Brownian process incrementing from t to $(t+1)$ with Δt infinitesimally small.

- $\Delta W \sim N(0, \Delta t)$, which implies $E[\Delta W] = 0$ and $E[(\Delta W)^2] = \Delta t$
- As $\Delta t \to 0$;
  a. $(\Delta t)^2$ is of higher order than Δt
  b. $\Delta t\Delta W$ is of higher order than Δt
  c. $(\Delta W)^2$ behaves like Δt

When taking the limit $\Delta t \to 0$, the terms $(\Delta t)^2$ and $\Delta t\Delta W$ vanish faster then Δt and $(\Delta W)^2$. We can effectively say

$$(\Delta X)^2 \approx \sigma^2(\Delta W)^2$$

For the limit $(\Delta W)^2 = \Delta t$, substituting these back to the Taylor expansion

$$\Delta Y \approx \frac{\partial f}{\partial x}(\mu\Delta t + \sigma\Delta W)dt + \frac{\partial f}{\partial t}\Delta t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}\sigma^2(\partial W)^2$$

We take the limits as $\Delta t \to 0$ and dY_t as the limit of Δt.

The term $\frac{\partial f}{\partial x}\sigma\Delta W$ corresponds to the Ito integral part of dY_t.

The terms multiplied by Δt will form the drift part of dY_t.

When we use the property that $(dW_t)^2$ contributes as Δt term, the expression becomes

$$\Delta Y \approx \left(\frac{\partial f}{\partial x}\mu + \frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}\sigma^2\right)\Delta t + \left(\frac{\partial f}{\partial x}\sigma\right)\Delta W$$

Converting the above expression to differential form

$$dY_t \approx \left(\frac{\partial f}{\partial x}\mu(X_t, t) + \frac{\partial f}{\partial t}(X_t, t) + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(X_t, t)\sigma^2(X_t, t)\right)dt + \left(\frac{\partial f}{\partial x}(X_t, t)\sigma(X_t, t)\right)dW \tag{8}$$

The above expression is Ito's lemma and highlights how $\frac{1}{2}\sigma^2\frac{\partial^2 f}{\partial x^2}$ arises from the non-zero quadratic variation of the Brownian process captured by $(\Delta W)^2 \approx \Delta t$ property when applying the Ito integral.

Example 1

Consider a stochastic process

$$f(X_t, t) = tX_t^2$$

We derive the Ito's lemma using the properties of the Ito integral and Taylor series expansion.

The X_t process is defined by the following SDE:

$$dX_t = \mu(X_t, t)dt + \sigma(X_t, t)dW_t$$

where $\mu = 0.05x$ and $\sigma = 0.2x$.

We treat the drift and diffusion coefficients as constants μ and σ respectively. We are interested in the behavior of a certain function $Y_t = f(X_t, t)$.

In practical application, we would want to know how returns (Y_t) of a trading strategy will evolve given a price function (X_t) as a stochastic process parameter.

We refer to the second order Taylor series expansion

$$dY_t \approx \frac{\partial f(X_t, t)}{\partial t}dt + \frac{\partial f(X_t, t)}{\partial x}dX_t + \frac{1}{2}\sigma^2\frac{\partial^2 f(X_t, t)}{\partial x^2}(dW_t)^2$$

Solve for the PDEs:

$$\frac{\partial}{\partial t}f(X_t, t) = X^2$$
$$\frac{\partial}{\partial x}f(X_t, t) = 2tX$$
$$\frac{\partial^2}{\partial x^2}f(X_t, t) = 2t$$

Substituting $dX_t = \big(0.05X\big) \cdot dt + \big(0.2X\big) \cdot dW_t$, and the PDEs into dY_t

$$dY_t \approx x^2 dt + 2tX \cdot \big((0.05X)dt + (0.2X)dW_t\big) + \frac{1}{2}(0.2X)^2 \cdot (2t) \cdot (dW_t)^2$$

Gives us the returns of the trading strategy as:

$$dY_t \approx \big(0.14tX^2 + X^2\big)\,dt + \big(0.4tX^2\big)\,dW_t$$

Fig. (3) shows the graph of the underlying asset price and the returns of the strategy trading strategy.

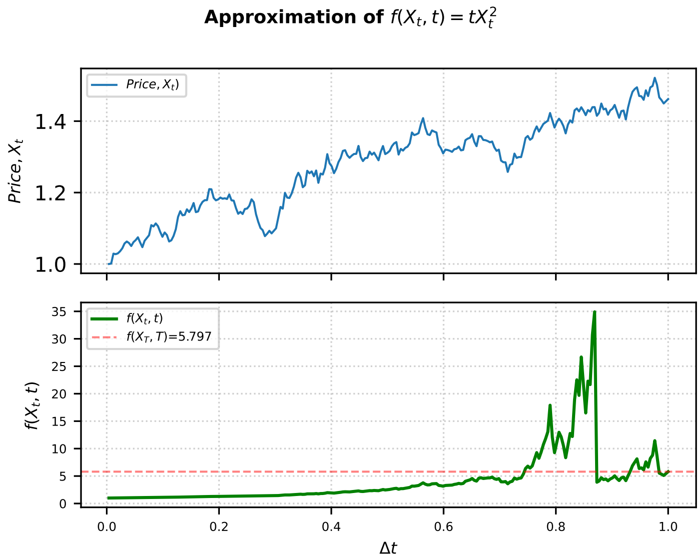

Figure 3: Approximation of $f(X_t) = tX_t^2$

Example 2

Consider the SDE for asset price S_t:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

We want to find the SDE for the log price, Y_t, expressed as

$$f(S_t) = \ln(S_t)$$

Ito's lemma states:

$$df(S_t) = \frac{\partial f}{\partial S} dS_t + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}(dS_t)^2$$

Solve the PDE's

$$\frac{\partial f}{\partial S} = \frac{1}{S_t}$$
$$\frac{\partial^2 f}{\partial S^2} = -\frac{1}{S_t^2}$$

Solve for $(dS_t)^2$ term of Ito's lemma

$$(dS_t)^2 = (\mu S_t dt + \sigma S_t dW_t)^2$$

Expanding this out algebraically

$$(dS_t)^2 = \mu^2 S_t^2 (dt)^2 + 2\mu\sigma S_t^2 (dt)(dW_t) + \sigma^2 S_t^2 (dW_t)^2$$

Now we apply the formal multiplication rules of stochastic calculus as $dt \to 0$:

- $(dt)^2 \to 0$ (it shrinks to zero much faster than dt)
- $(dt)(dW_t) \to 0$
- $(dW_t)^2 \to dt$ (the fundamental property of variance in Brownian motion)

Applying these rules makes the first two terms vanish entirely

$$(dS_t)^2 = \sigma^2 S_t^2 dt$$

Substitute the PDEs and the squared identity into the general Ito's lemma equation:

$$df(S_t) = \left(\frac{1}{S_t}\right) dS_t + \frac{1}{2}\left(-\frac{1}{S_t^2}\right)(\sigma^2 S_t^2 dt)$$

Substitute the actual definition of dS_t $(\mu S_t dt + \sigma S_t dW_t)$ into the first term:

$$df(S_t) = \frac{1}{S_t}(\mu S_t dt + \sigma S_t dW_t) - \frac{1}{2}\sigma^2 dt$$

Distribute the $\frac{1}{S_t}$ across the parentheses:

$$df(S_t) = \mu dt + \sigma dW_t - \frac{1}{2}\sigma^2 dt$$

Finally, group the common dt terms together to get the final SDE for $Y_t = \ln(S_t)$:

$$dY_t = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t$$

We simulate both the underlying asset price S_t and its analytical log-transformed Y_t as shown in Fig.(4)

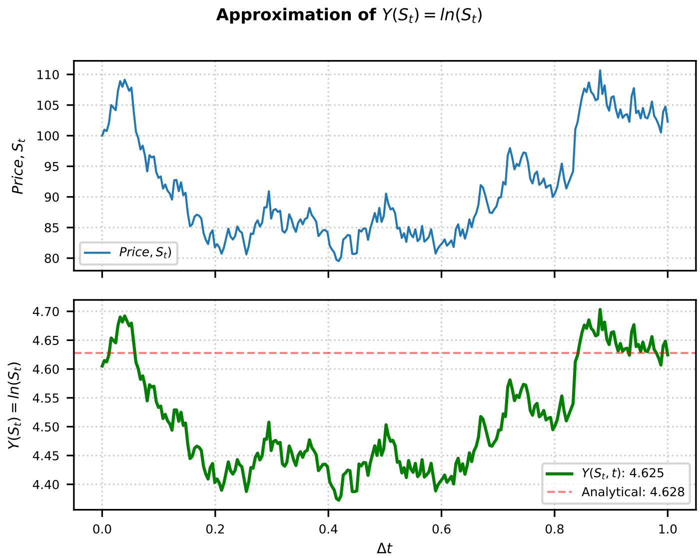

Figure 4: Approximation of $Y(S_t) = ln(S_t)$

5 Discrete Time

From the simple trading strategy(Jonathan Goodman,), let S_t be the price of the asset that can be bought or sold at any time t for $t = 0, 1, \cdots$ where $t = 0$ and the initial S_0 are known, but future fluctuations Y_k are unknown. We model the price change from time t to n as a random variable Y_k, then the price process S_n is a random walk

$$S_n = \sum_{t=0}^{n-1} Y_k$$

which is also equivalent to

$$Y_n = S_{n+1} - S_n$$

The price changes by an amount of Y_n going to time $n+1$ where the gain or loss is $\alpha_n \cdot Y_n$. We assume that trading in this model is free $(\alpha_{t+1} \neq \alpha_t)$ and that the trader is a price taker where any number of α_t is allowed (i.e., the market has an infinite supply of shares) and that buying and selling shares do not affect the price. The value of the position at the beginning of the period k is $\alpha_k S_k$, which is α_k shares, each of which has the price of S_k. After the period k, the price has changed to $S_{k+1} = S_k + Y_k$. The total gain up to time n is:

$$X_n = \sum_{k=0}^{n-1} \alpha_k (S_{k+1} - S_k) \tag{9}$$

The strategy is non-anticipating if α_k is only determined by $(Y_0, \cdots, Y_{k-1})$. In this way, we put the bet on Y_k without knowing what Y_k will be. A strategy starts with an initial bet α_0 and must not depend on the returns of S_k at all. After that, the bets are $\alpha_1(Y_0)$, $\alpha_2(Y_0, Y_1)$ and so on up to $\alpha_k(Y_0, Y_1, \cdots, Y_{k-1})$.

This is decision-making under uncertainty, where we use the probability of the future. This may include an uncertain forecast, at time k, Y_k is a random variable. A non-anticipating trading strategy α_k produces a sequence of random variables X_n. The questions are as follows

1. What random variables can be achieved in this way?
2. Which of the random paths is optimal?

A particularly important case is when the price process S_t is a simple random walk. This means that steps S_k satisfy these three properties:

$$\mathbb{E}[Y_k] = 0, \quad Var(Y_k) = \sigma^2, \quad \text{where all } Y_k \text{ are i.i.d.} \tag{10}$$

Definition 5.1 (Martingale) . *A martingale is a sequence of random variables where the expected value given all past information is exactly equal to the most recent value. A sequence $\{X_n\}$ is a martingale if it satisfies the following conditions:*

- *The absolute expected value is finite $\mathbb{E}\big[|X_n|\big] < \infty$.*
- *The value of X_t is determined solely by the information available up to time t.*
- *The conditional expectation of the next value X_{t+1}, given all previous values, is the current value X_t.*

Definition 5.2 (Doob Martingale) *A Doob martingale is a stochastic process that provides the best estimate of a random variable as more information becomes available over time. It is a sequence of evolving approximations. Formally, given a complex random variable Y and a sequence of information snapshots, a filtration F_t, the Doob martingale X_t is defined as the conditional expectation of Y given all information up to time t.*

$$X_t = \mathbb{E}[X_{t+1}|F_t] = \mathbb{E}[Y|F_t]$$

Unlike random walks that can fluctuate to infinity, a Doob martingale always converges as $t \to \infty$ provided that the original Y is integrable. Every Doob martingale is uniformly integrable and will not have extreme explosive behavior.

Any non-anticipating strategy applied to a martingale produces another martingale. One cannot manufacture a positive expected value by strategizing from a non-zero expected return process. We call the sequence $\{S_0, S_1, \cdots, S_k, \cdots, S_n\}$ a discrete path. The time variable k is discrete, but the value of S_k could be continuous or discrete. The process is a martingale if the conditional expectation of the future value is equal to the present value.

$$\mathbb{E}\big[S_{n+1}\big|(S_0, \cdots, S_n) = (s_0, \cdots, s_n)\big] = s_n$$

The martingale property is expressed in terms of increments and decreases in the form of $S_{k+1} = S_k + Z_k$ and $Z_k = S_{k+1} - S_k$ where Z_k is conditional on knowing the value of S_n:

$$\mathbb{E}\big[Z_n\big|(S_0, \cdots, S_n) = (s_0, \cdots, s_n)\big] = 0 \tag{11}$$

The martingale and Markov properties seem similar, but they are not the same. A Markov process does not have to be a martingale. In fact, the expression (2) implies that an SDE process is a martingale only if the infinitesimal mean is equal to zero. Moreover, the distribution of the martingale would depend on s_n only. For example, if the numbers Y_k satisfy the properties of unbiased random walk(10), then the following expression defines a martingale

$$S_{n+1} = S_n + S_{n-1} \cdot Y_n + S_{n-2} \cdot (Y_n^2 - \sigma^2)$$

The last two terms on the right have an expected value of zero, even given the values of $S_{n-1} = s_{n-1}$ and $S_{n-2} = s_{n-2}$. The distribution of the increments may depend on the whole path, but regardless of which path is taken up to time n, the increments will have a mean equal to zero. However, note that the mean of the unconditional increments may be zero, although the process is not a martingale.

The unconditional expected value is $\mathbb{E}[S_{n+1} - S_n]$. This can be zero even though the conditional expected value given by (11) is not all zero. On the other hand, if S_N is a martingale, then the unconditional means are zero due to the stochastic property stating that unconditional mean is the mean of the conditional means.

The Doob theorem for this martingale says that if S_n is a martingale and X_n results from the sequence of S_k by a strategy of the form (9) and if the strategy weights of α_k are non-anticipating, then X_n is a martingale.

As a simple proof, consider a strategy where α_n is a function of all that has come before $\alpha_n = \alpha(s_1, \cdots, s_n, n)$. The increments of the X process is $Z_n = Y_n\alpha(S_1, \cdots, S_n, n)$. Given the S path up to n, the conditional expectation is

$$\mathbb{E}\big[Y_n \cdot \alpha(S_1, \cdots, S_n, n)|S_1 = s_1, \cdots, S_n = s_n\big] = \\ \mathbb{E}\big[Y_n\big|S_1 = s_1, \cdots, S_n = s_n\big] \cdot \alpha(s_1, \cdots, s_n) = 0$$

The constant $\alpha(s_1, \cdots, s_n)$ is just a number, not a random variable and must be outside the expectation. Active strategies cannot turn a martingale into an expected profit, but they can give a lower risk with the same expected return. The Ito isometry for this situation is

$$\mathbb{E}[X_n^2] = \sigma^2 \sum_{k-0}^{n-1} \mathbb{E}[\alpha_k^2] \tag{12}$$

In this theorem, the sequence $\{X_n\}$ is constructed from the sequence $\{S_n\}$ by any non-anticipating strategy (9). Proof. We write X_n as a double index summation:

$$X_n = \sum_{k=1}^{n} \alpha_k Y_k = \sum_{j=1}^{n} \alpha_j Y_j$$

The coefficients α_k could be complicated functions $Y_1, \cdots, Y_{k-1}$. The proof starts with the trick of using both sums to express the square and then interpreting the product of sums as a sum of products.

$$\begin{aligned}
\mathbb{E}[X_n^2] &= \mathbb{E}\left[\left(\sum_{k=1}^{n} \alpha_k Y_k\right)\left(\sum_{j=1}^{n} \alpha_j Y_j\right)\right] \\
&= \mathbb{E}\left[\sum_{k=1}^{n}\sum_{j=1}^{n} \alpha_k Y_k \cdot \alpha_j Y_j\right] \\
&= \sum_{k=1}^{n}\sum_{j=1}^{n} \mathbb{E}\left[\alpha_k Y_k \cdot \alpha_j Y_j\right] \\
&= \sum_{k=1}^{n}\sum_{j=1}^{n} \alpha_k \alpha_j \mathbb{E}\left[Y_k Y_j\right]
\end{aligned}$$

Moreover, we need to know the properties of Y_k and α_k

Case 1. Y_k are independent and have zero mean and α_k are constants.

if Y_k are independent random variables with $\mathbb{E}[Y_k] = 0$ for all k, then:
If $k \neq j$; $\mathbb{E}[Y_k Y_j] = \mathbb{E}[Y_k]\mathbb{E}[Y_j] = 0 \cdot 0 = 0$
If $k = j$; $\mathbb{E}[Y_k Y_k] = \mathbb{E}[Y_k^2] = Var(Y_k) + (E[Y_k])^2 = Var(Y_k) + 0^2 = Var(Y_k)$

In this case, the double summation simplifies because all items where $k \neq j$ become zero, we need only to consider the terms where $k = j$:

$$\mathbb{E}[X_n^2] = \sum_{k=1}^{n} \alpha_k \alpha_k \mathbb{E}[Y_k Y_k] = \sum_{k=1}^{n} \alpha_k^2 \mathbb{E}[Y_k^2]$$

if Y_k are also identically distributed with variance σ_Y^2 and mean 0, then $\mathbb{E}[Y_k^2] = \sigma_Y^2$ for all k

$$\mathbb{E}[X_n^2] = \sum_{k=1}^{n} \alpha_k^2 \sigma_Y^2 = \sigma_Y^2 \sum_{k=1}^{n} \alpha_k^2$$

Case 2. Y_k are independent and identically distributed with mean μ_Y and variance σ_y^2, and α_k are constants. if Y_k are independent random variables with $\mathbb{E}[Y_k] = \mu_Y$ and $Var(Y_k) = \sigma_Y^2$, then $E[Y_k^2] = Var(Y_k) + (E[Y_k])^2 = \sigma_Y^2 + \mu_Y^2$. For $k \neq j$; $E[Y_k Y_j] = E[Y_k]E[Y_j] = \mu_y \cdot \mu_y = \mu_Y^2$ so

$$\mathbb{E}[X_n^2] = \sum_{k=1}^{n} \sum_{j=1}^{n} \alpha_k \alpha_j \mathbb{E}[Y_k Y_j]$$

We can split the sum into diagonal ($k = j$) and off diagonal ($k \neq j$) terms.

$$E[X_n^2] = \sum_{k=1}^{n} \alpha_k^2 E[Y_k^2] + \sum_{k=1}^{n} \sum_{j=1, j\neq k}^{n} \alpha_k \alpha_j E[Y_k Y_j]$$

$$E[X_n^2] = \alpha_k^2(\sigma_k^2 + \mu_k^2) + \sum_{k=1}^{n} \sum_{j=1, j\neq k}^{k} \alpha_k \alpha_j \mu_Y^2$$

This can be rewritten using the fact $(\sum_{k=1}^{n} \alpha_k)^2 = \sum_{k=1}^{n} \alpha_k^2 + \sum_{k=1}^{n} \sum_{j=1, j\neq k}^{n} \alpha_k \alpha_j$. Let $A = \sum_{k=1} \alpha_k$. Then, $A^2 = \sum_{k=1}^{n} \alpha_k^2 + \sum_{j \neq k} \alpha_k \alpha_j \mu_Y^2$.

$$E[X_n^2] = B(\mu_k^2 + \sigma_k^2) + (A^2 - B)\mu_Y^2$$
$$E[X_n^2] = B\sigma_k^2 + B\mu_Y^2 + A^2\mu_Y^2 - B\mu_Y^2$$
$$E[X_n^2] = B\sigma_Y^2 + A^2\mu_Y^2$$

Without the assumption about Y_k , like independence or mean, the general form solution is

$$E[X_n^2] = \sum_{k=1}^{n} \sum_{j=1}^{n} \alpha_k \alpha_j E[Y_k Y_j]$$

This formula represents the expected value of X_n^2. The name isometry comes from the mathematical transformation that does not change in size. In this case, one of the objects is the sequence for the random variable μ. The size of the sequence is given by the sum of expected squares. The other object is the random variable X_n, whose size is measured by $E[X_k^n]$.

Ito's lemma for the discrete-time integral is essentially the stochastic version of the chain rule. For a function $f(W_t)$, we use the standard Taylor expansion on $f(W_{t_{n+1}})$ around W_{t_n}.

$$\begin{aligned} f(W_{t_{n+1}} - f(W_{t_n}) &\approx f'(W_{t_n})\Delta W_{t_n} \\ &\quad + \frac{1}{2} f"(W_{t_n})(\Delta W_{t_n})^2 + \cdots \end{aligned}$$

Note that in continuous time, the squared differential $(dW_t)^2$ is deterministically equal to dt. In contrast to discrete time, $(dW_t)^2$ is a random variable. While it is expected values is Δt, it is not perfectly equal to Δt at every step. Therefore, the discrete Ito formula carries a localized error

$$\begin{aligned} \Delta f(W_{t_n}) &\approx f'(W_{t_n})\Delta W_{t_n} + \frac{1}{2} f"(W_{t_n})\Delta t \\ &\quad + \frac{1}{2} f"(W_{t_n})[(\Delta W t_n)^2 - \Delta t] \end{aligned}$$

The term $[(\Delta W_{t_n})^2 - \Delta t]$ is a mean zero noise term. As $\Delta t \to 0$, the sum of these errors vanishes. This is how we get the clean deterministic $\frac{1}{2} f''(W_{t_n})dt$ term in the continuous calculus.

If a continuous process is described as

$$dX_t = \mu(X_t)dt + \sigma(X_t)dW_t$$

the discrete time simulation scheme is

$$X_{t_{n+1}} = X_{t_n} + \mu(X_{t_n})\Delta t + \sigma(X_{t_n})\Delta W_{t_n}$$

The continuous increment of ΔW_{t_n} in the third term is approximated as $\Delta W_k = W_{t_{k=1}} - W_{t_k}$. Moreover, the integral of this continuous increment becomes the summation of $\int X_t dW_t \approx \sum X_{t_k} \Delta W_k$.

The discrete time ΔW_k is approximated using independent and normally distributed random variables scaled by Δt

$$\Delta W_k \sim \mathbb{N}(0, \Delta t) \text{ or } \Delta W_k = \mathcal{Z}_k \sqrt{\Delta t}$$

where $\mathcal{Z}_k$ is a standard normal random variable $\mathcal{N}(0, 1)$.

In Fig. 5, we keep the non-anticipating evaluation of μ and σ to maintain the martingale property of the random process. This discretization evaluates the coefficient S_{t-1} at the beginning of the time interval. The random shock is multiplied by $\sqrt{\Delta t}$, not Δt. This preserves the physical property of Brownian motion, where variance scales linearly with time, $\mathrm{Var}(W_t) = t$.

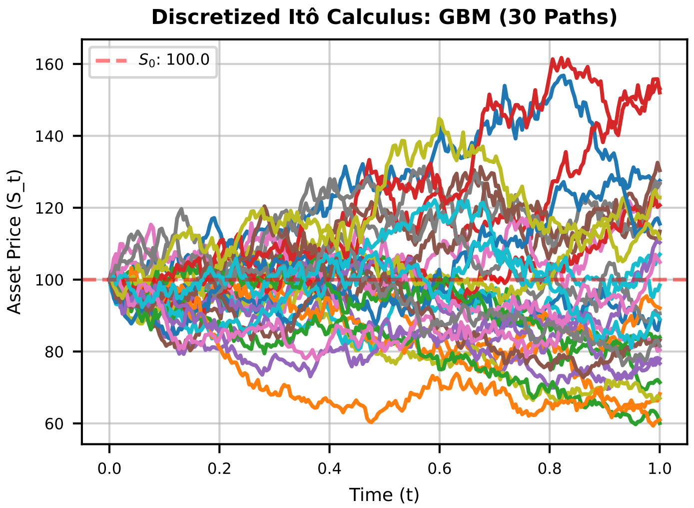

Figure 5: Ito Calculus Discrete

6 Continuous Time

The Itô integral arises as the continuous-time limit of the discrete trading-gain sum introduced in (13). It captures the dynamics of a setting in which the asset price evolves continuously, while the trading strategy is updated at a comparable (potentially very high) frequency.

Let $W_{t_{t\geq 0}}$ be a standard Brownian motion, interpreted here as the (idealized) price process of the asset at time t. Fix a small time increment $\Delta t > 0$, which represents the temporal resolution at which the price is observed and the trading strategy can be adjusted. We discretize time into intervals of length Δt and denote the grid points by $t_k = k\Delta t$, for $k \in \mathbb{N}$.

At each time t_k, the strategy chooses a position size α_{t_k} in the asset, where α_{t_k} may depend on the information available up to time t_k. Over the subsequent interval $[t_k, t_{k+1}]$, the price increment is given by $W_{t_{k+1}} - W_{t_k}$. Consequently, the profit or loss realized over this period is

$$\alpha_{t_k}(W_{t_{k+1}} - W_{t_k})$$

The total profit up to $t + \Delta_t$ is

$$X_t^{\Delta_t} = \sum_{t_k < t} \alpha_{t_k} \left(W_{t_{k+1}} - W_{t_k}\right) \tag{13}$$

The Ito integral with respect to the Brownian motion is the limit as $\Delta_t \to 0$

$$X_t = \int_0^t \alpha_s dW_s = \lim_{\Delta_t \to 0} \sum_{t_k < t} \alpha_{t_k} \left(W_{t_{k+1}} - W_{t_k}\right) \tag{14}$$

The continuous time limit $\Delta_t \to 0$ is like the continuous time limit of the Riemann integral

$$\int_0^t b_s ds = \lim_{\Delta_t \to 0} \sum_{t_k < t} b_{t_k} \Delta_t$$

The Riemann integral limit exists as long as b_t is a continuous function of t. The Ito integral limit (14) exists if α_t is continuous and non-anticipating martingale process. This means that there is a function $A(W_{[0,t]}, t)$ so that

$$\alpha_t = A(W_{[0,t]}, t)$$

For example, we could ignore the random part and take $\alpha_t = t^2$. The Ito integral would be a random variable

$$X_t = \int_0^t s^2 dW_s$$

Another example is $\alpha_t = W_t$. This is also non-anticipating since $1 + X_t$ depends on W_s for $s \leq t$

$$X_t = 1 + \int_0^t X_s dW_s$$

The limit of (14) is a sequence of random variable $X_t^{\Delta_t}$ converges to another random variable X_t. There are several kinds of convergence for random variables, including convergence in distribution, convergence in probability, and convergence almost surely. Convergence in distribution means that the probability distribution of the random paths $X_t^{\Delta_t}$ converges to the probability distribution of X_t. As a result, simulating the process $X_t^{\Delta_t}$ gives results similar to the theoretical process X_t which we cannot simulate exactly. Convergence in probability is the statement that for every $\epsilon > 0$

$$Pr\left(\left|X_t^{\Delta_t} - X_t\right| > \epsilon\right) \to 0 \quad \text{as} \quad \Delta_t \to 0$$

This would be natural if we are estimating X_t using $X_t^{\Delta t}$. Almost sure convergence says that with probability 1 the limit formula of (14) is true. Almost sure convergence is the hardest to prove theoretically and most difficult to check in computations. We will need a sequence of simulations with decreasing Δt.

For convergence in probability, a simulation of single Δt might suffice. The smaller Δt the less likely the results will be off by more than ϵ.

For almost sure convergence, it is unlikely ever to be wrong by more than ϵ as $\Delta t \to 0$. The Doob martingale theorem for Ito integral which states that if α_t is non-anticipating then X_t is a martingale. The definition of martingale in continuous time is that X_t is a martingale with respect to the Brownian motion path W if

$$\mathbb{E}\left[X_{t+s} | W_{[0,t]}\right] = X_t$$

where $W_{[0,t]}$ is the Brownian motion in the interval $[0, t]$ and X_t is a function of $W_{[0,t]}$ under the condition that X_t is an Ito integral with respect to W. This definition is a bit different in the discrete time for martingale. where X_n is the current position, andX_{n+1} is the next step. In contrast to continuous time, the next step is referred as all times for $s > t$. The proof of this version of Doob's theorem is just from the definitions and limits. The Ito integral is the continuous time limit $\Delta_t \to 0$ of the discrete time sum described in (13) and therefore, $X_t^{\Delta_t}$ is indeed a martingale. The complete mathematical proof is a little involved, because $X_t^{\Delta_t}$ is martingale only in the discrete times t_k. These get more finely spaced in the continuous time limit. It is a basic technical theorem in probability that if a discrete time martingale converges in this way, the limit is a continuous time martingale. The continuous time Ito isometry for this kind of Ito integral is

$$E[X_t^2] = \int_0^t E[a_s^2] ds \tag{15}$$

We prove this by using the discrete time Ito isometry theorem (12) and taking the continuous time limit. The parameter σ^2 in discrete time is the variance of the martingale difference. The martingale difference for $X_t^{\Delta_t}$ is $\Delta W_k = W_{t_{k+1}} - W_{t_k}$. Then, the variance is

$$\sigma^2 = Var(\Delta W_k) = E\left[\left(W_{t_k+\Delta t} - W_{t_k}\right)^2\right] = \Delta_t$$

Therefore, the discrete time in (12) implies that

$$E[(X_{t_n}^{\Delta t})^2] = \Delta t \sum_{k=0}^{n-t} E[\alpha_{t_k}^2]$$

in the continuous time limit, the right side sum converges to the Riemann integral on the right side of the continuous time (15) . Take note that the approximation in (13) requires ΔW_k to be in the future of t_k.

In Fig. (6), the continuous time integral $\int S_t dW_t$ represents the infinitesimal increment of the Brownian process. Given the SDE, $dS_t = \mu S_t dt + \sigma S_t dW_t$, we use Ito's lemma, which is derived from the second-order Taylor expansion and gives:

$$S_t = S_0 \cdot \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t\right) \tag{16}$$

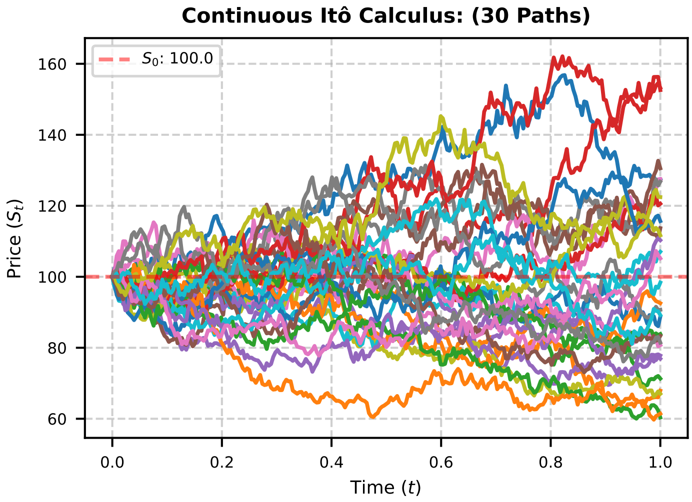

Figure 6: Ito Calculus Continuous

7 Conclusion

We examined the structure of a stochastic process and provided a detailed analysis of its construction and properties. Through numerical simulations, we demonstrated the importance of Ito's integral and Ito's lemma in solving stochastic differential equations and why the standard differentiation fails to capture the real data from a stochastic process.

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